

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch : Electronics and Telecommunications Semester : III Subject Code & Name: Digital Logic Design BTEXC305 Max Marks: 60 Date: Duration: 3 Hr.		
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.		
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	Explain Parity generators/checkers	L1/CO1	6
B)	Write a note on Digital Comparator	L1/CO2	6
C)	Solve using K-map $F(A,B,C,D) = \sum(m_0, m_1, m_2, m_4, m_5, m_6, m_8, m_9, m_{12}, m_{13}, m_{14})$	L1/CO3	6
Q.2	Solve Any Two of the following.		
A)	Write a note on 1 Bit Memory Cell, explain J-K flip flop.	L2/CO1	8
B)	Discuss applications of flip flops, explain ring counter in detail.	L2/CO2	8
C)	Define of lock out, Clock Skew, and Clock jitter.	L2/CO3	8
Q. 3	Solve Any One of the following.)		
A)	With example explain State diagram, State table, State reduction, State assignment	L3/CO1	10
B)	Explain finite state machine implementation	L3/CO2	10
Q.4	Solve Any Two of the following.		
A)	Describe any three Characteristics of digital ICs.	L1/CO5	6
B)	Compare TTL logic and CMOS logic	L1/CO4	6
C)	Explain CMOS inverter, NAND, NOR gates.	L1/CO5	6
Q. 5	Solve Any One of the following		
A)	Compare architecture of FPGA and CPLD	L1/CO4	10
B)	Classify and write characteristics of memories.	L1/CO6	10
	*** End ***		

The grid and the borders of the table will be hidden before final printing.